ALLAN ANILSON

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EDUCATION

- **TKM College of Engineering, Kerala, India** 2021 - 2025
  Bachelor of Technology (B.Tech) in Computer Science and Engineering

EXPERIENCE

- **Guidehouse – AI Engineer** August, 2025 – Present
  Thiruvanthapuram, Kerala, India
  - Engineered a full-stack Employee Relations (ER) platform using Next.js and FastAPI, automating case ingestion from the MS Graph API into a PostgreSQL database; integrated LLMs for automated email summarization and multi-tier categorization, providing a real-time dashboard for case management. Helped reduce average case triage time by approximately 70% and manual processing effort by 60%.
  - Architected and deployed a cloud-native SOP search platform for healthcare GMS, building an end-to-end RAG pipeline leveraging Azure Blob Storage and Azure AI Search, integrated through FastAPI and Next.js to provide low-latency, source-based compliance answers.
  - Optimized enterprise-wide document access by building a Copilot Studio agent leveraging 27+ SharePoint sites for real-time RAG, engineering prompt orchestration, and scaling the solution to 5000+ users through Microsoft Teams.
- **Sedai – DevOps Engineer Intern** August, 2024 – March, 2025
  California, United States
  - Developed a K-Means clustering algorithm to classify Kubernetes worker nodes based on real-time CPU utilization, improving cluster resource utilization by approximately 25%.
  - Designed safe workload eviction for Deployments, StatefulSets, and Jobs while preserving application availability and improving cluster packing efficiency.
  - Automated workload migration using Kubernetes taints, cordon, drain, and pod eviction, enabling zero-downtime redistribution across the cluster.

PROJECTS

- **WorkspaceAI – Multi-Tenant Knowledge Management Platform**
  - **Developed**: a production-grade multi-tenant collaboration platform using FastAPI, Next.js, PostgreSQL, AWS, Docker, JWT, and RBAC, delivering secure multi-tenant architecture and scalable REST APIs.
  - **Implemented**: AI-powered knowledge discovery using RAG, semantic search, embeddings, document summarization, and AI agents, reducing document discovery time by 40%.
  - **Deployed**: the platform on AWS using Docker and Terraform, enabling scalable, cloud-native AI workloads with enterprise-grade reliability.
- **ResearchGPT – Multi-Agent AI Research Platform**
  - **Developed**: a multi-agent AI research platform using FastAPI, LangGraph, PostgreSQL, pgvector, Redis, and Docker, orchestrating Planner, Retriever, Critic, Fact Checker, and Citation Verifier agents.
  - **Implemented**: a hybrid RAG pipeline combining Tavily, semantic vector search, and parallel agent workflows, improving factual accuracy by 35% while reducing hallucinations.
  - **Deployed**: the platform on AWS using Docker and Terraform, integrating LangSmith, Prometheus, and Grafana for observability, workflow tracing, and performance monitoring.

CERTIFICATIONS

- AWS Certified Solutions Architect - Associate - Amazon Web Services
- AWS Certified AI Practitioner - Amazon Web Services
- Machine Learning Specialization Offered by DeepLearning.AI & Stanford University - Coursera
- NPTEL - Python for Data Science - IIT Madras
- Linux Mastery: Master the Linux Command Line - Udemy

SKILLS

- **Programming**: Python, TypeScript, JavaScript, SQL, Java, Bash
- **AI & GenAI**: LLMs, RAG, AI Agents, LangChain, LangGraph, Prompt Engineering, Context Engineering, Embeddings, Semantic Search, Vector Search, Evaluation, Tavily
- **Backend**: FastAPI, SQLAlchemy, Django, REST APIs, GraphQL, PostgreSQL, MySQL, Redis
- **Cloud**: AWS, Microsoft Azure, Docker, Kubernetes, Terraform, Linux
- **Tools**: Git, GitHub Actions, Prometheus, Grafana